Muhammad Aswab Khalil

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LinkedIn | GitHub | **Portfolio**

PROFESSIONAL SUMMARY

Front-end-focused Software Engineer with over a year of professional experience as a full-stack and Flutter developer (CarMandi, CHORE Robotics). Builds responsive, component-based user interfaces with React.js, JavaScript/TypeScript, HTML5, CSS3 and Tailwind CSS, backed by Node.js/Express and REST APIs. Shipped a production-ready Flutter app with offline-first architecture, plus full-stack platforms with authentication, media pipelines and analytics dashboards. Strong foundation in applied computer vision and AI. Comfortable owning a project end-to-end, from UI and architecture to deployment.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Dart, Python, C++, Java, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Design, Component-Based UI, State Management (Provider), Chart.js, REST API Integration
Mobile Development: Flutter, Provider, GoRouter, Camera Integration, Image Picker, Offline Storage, Deep Links, Push Notifications
Backend & Databases: Node.js, Express.js, JWT Authentication, MongoDB
AI / Computer Vision: OpenCV, NumPy, OCR, Image Processing, Feature Extraction, ResNet, SIFT, Bag of Visual Words (BoVW), Epipolar Geometry, 3D Reconstruction
Tools & Platforms: Git, GitHub, Docker, Linux, Firebase, AWS (EC2), Cloudflare R2, Cloudinary, Vercel

EXPERIENCE

Full-Stack Developer | CHORE Robotics | Jan 2026 – Present
Website: chorerobots.ai
- Built and maintained the company website, backend APIs, and database integrations.
- Developed an admin dashboard with media management, form submission tracking, and account-based permissions.
- Delivered features end-to-end, including testing, Cloudflare R2 storage integration, and AWS EC2 deployment.

Full-Stack & Flutter Developer | CarMandi | Nov 2024 – Jan 2026
Website: carmandi.pk | App: Google Play | App Store
- Built and maintained Flutter Android/iOS features, web interfaces, and backend services.
- Integrated APIs, deep links, push notifications, and inspection booking workflows.
- Resolved production issues and managed deployments through GitHub and AWS.

PROJECTS

Vehicle Inspection Mobile Application | Flutter, Dart, Provider, GoRouter
- Developed a production-ready Flutter application supporting structured, multi-step vehicle inspection workflows.
- Built camera-based damage capture and mapping features to document vehicle condition in the field.
- Implemented offline-first local storage with automatic background synchronization for reliable data capture without connectivity.
- Migrated the application from a low-code prototype to a fully source-controlled, maintainable Flutter codebase.

GitHub: github.com/aswab007-ops/car-inspection-app

RozGo — Gig Income Tracker | React, Express.js, MongoDB, JWT, Cloudinary
- Built a full-stack MERN application for freelancers to track and manage gig-based income, with a React front end.
- Implemented JWT authentication with email verification and Cloudinary-based file uploads.
- Developed dashboard analytics (Chart.js), PDF export, and an admin review workflow for submission approval.
- Deployed the application on Vercel for public access.

GitHub: github.com/aswab007-ops/RozGo

Camera Calibration & 3D Reconstruction | Python, OpenCV, NumPy
- Implemented projection, fundamental, and essential matrix estimation for multi-view geometry.
- Applied epipolar geometry constraints and 3D triangulation to reconstruct scene structure from image pairs.

GitHub: github.com/aswab007-ops/cameramodel

Scene Recognition using BoVW & ResNet | Python, OpenCV, Scikit-learn
- Compared classical Bag of Visual Words (SIFT + K-Means) against deep ResNet features for scene classification.
- Trained and evaluated KNN and SVM classifiers, benchmarking accuracy across both feature representations.

GitHub: github.com/aswab007-ops/scene-recognition-using-bow-and-resnet

Additional Projects: Freelancers Network (MERN platform with JWT auth, job filtering & bookmarking), Custom Canny Edge Detector, Lane Detection System (HSV masking, Hough Transform), Person Removal via Change Detection, Emoji Detector AI (OCR-based) — see GitHub for details.

EDUCATION

Information Technology University (ITU), Lahore
Bachelor of Science in Artificial Intelligence (BSAI) | CGPA: 2.78